

Small Language Model Challenge

DASE7506 Project 1 - Reproducible final report

Frozen full-test result: 1.605136467 BPB

Initial baseline: 2.102014912 BPB | Absolute gain: 0.496878446 BPB | Relative reduction: 23.64%

Abstract

This project trains a causal language model from random initialization on the supplied WikiText-2 training split under strict CPU inference limits. The final model combines rotary position embeddings, SwiGLU feed-forward layers, pre-normalized RMSNorm, tied embeddings, width 256, depth 6, and training-time dropout 0.10. Architecture changes were first compared at equal processed-target budgets; capacity, regularization, and duration were then selected only on validation. The frozen 5,247,744-parameter model achieves 1.605136467 full-test bits per byte (BPB), compared with 2.102014912 for the initial course baseline. A three-seed exact-recipe replication gives 1.579745 mean validation BPB with 0.003591 sample standard deviation. The final predictor passes all resource limits at 3.485x baseline CPU time, 1.972 GB peak RSS, and 21.14 MB of core inference assets.

Key outcomes

Measure	Baseline	Frozen model	Status
Full-test BPB	2.102014912	1.605136467	Lower by 0.496878446
Parameters	1,088,256	5,247,744	4.82x
CPU scoring time	12.384 s	43.163 s	3.485x; pass
Peak RSS	-	1.837 GiB	Below 4 GiB
Inference assets	-	20.16 MiB	Below 64 MiB

Scope and integrity

All training uses only the supplied training text. Model selection uses validation only. The method and exact hashes were frozen before the full test was scored. No test-driven tuning, pretrained weights, external text, cached answers, cross-window state, or evaluation-time network access is used.

1. Task and evaluation contract

The objective is to minimize bits per raw UTF-8 byte on the supplied WikiText-2 benchmark while retaining causal, independent 256-token evaluation windows. The fixed BPE tokenizer has 2,048 tokens. Each prediction can depend only on its observed prefix, and temporary state must reset across windows and examples.

Constraint	Required	Final evidence
Development data	Training text only; select on validation	All search and checkpoint selection used validation
Scoring	CPU FP32, fixed evaluator	Protocol 7506-mp1-wt2-v2; hashes frozen
CPU time	No more than 5x baseline	3.485x median over 3 fresh-process repeats
Peak RAM	No more than 4 GiB	1.836 GiB maximum peak RSS
Inference assets	No more than 64 MiB	20.16 MiB core assets
Test use	Freeze before testing	Pre-test manifest written before test evaluation

2. Final model

The final predictor is a decoder-only Transformer with four attention heads, model width 256, six blocks, and context length 256. It contains 5,247,744 trainable parameters. The output embedding is tied to the token embedding. Training uses dropout 0.10; dropout is disabled by evaluation mode and therefore does not add inference work.

Component	Choice	Rationale
Position encoding	Rotary position embedding (RoPE)	Strongest isolated architecture gain in the short screen
Feed-forward	SwiGLU	Gated nonlinearity; retained with RoPE after ablation
Normalization	Pre-normalized RMSNorm	Stable residual path and inexpensive normalization
Biases	Disabled in selected linear layers	Compact implementation; small incremental effect
Regularization	Dropout 0.10	Best matched 3,600-step validation result
Training	AdamW, cosine decay, 4,800 updates	Longer fresh schedule passed the preregistered threshold

The model implements both `forward(ids)` for training logits and `predict_log_probs(ids)` for finite normalized log probabilities. Contract tests check causality, normalized probabilities, example independence, window reset behavior, gradients, parameter counts, and reproducible resume.

3. Experimental protocol

Experiments were staged to separate mechanism evidence from later score optimization. Comparisons within each decision used fixed seeds, target counts, optimizer settings, and complete validation passes. Endpoints were kept distinct from validation-selected checkpoints. Every formal artifact records source, configuration, checkpoint, tokenizer, evaluator, seed, targets, and timing hashes.

Stage	Question	Decision rule	Outcome
1-2	Which mechanism helps at equal targets?	1,200 updates; matched seed and budget	RoPE dominates the isolated gain
3	Does the simplified mechanism replicate?	Seeds 17/23/42; equal-budget long comparison	RoPE+SwiGLU is robust; modern bundle kept for capacity
4-5	How much capacity fits the limits?	Short screen, then repeated CPU gate	Width 256/depth 6 selected
6-7	How much dropout?	Matched 3,600-step screen; bounded strength follow-up	0.10 selected; 0.15/0.20 rejected
8	Does longer training help?	Fresh 4,800-step run; minimum 0.002 BPB gain	Gain 0.022153; adopt
9	Does the final recipe replicate?	Seeds 23/42 with no recipe changes	Three-seed mean 1.579745
10	Does the frozen candidate meet resources?	Three alternating fresh-process repeats	All limits pass

Fair comparisons

The initial architecture screen processes 9,830,400 training targets per run. The later 4,800-step endpoint comparison processes 39,321,600 targets for every model. A completed short cosine run is never extended and relabeled as a fresh long schedule; long runs start from random initialization with the final duration planned in advance.

Checkpoint selection

Validation is evaluated every 300 steps in the final recipe. The selected checkpoint for each seed is independently rescored on CPU FP32. For all three final seeds, the best checkpoint is the 4,800-step endpoint. The prospective final checkpoint is the lowest CPU FP32 validation result across the three predeclared seeds, and this cross-seed validation selection is disclosed rather than presented as an unbiased replication mean.

Search cost

The two new Stage-9 replications process 78,643,200 targets and require 741.64 seconds of recorded GPU training time (836.81 process seconds). The selected seed-17 final run processes 39,321,600 targets in 373.25 GPU training seconds. Earlier failed and negative screens remain in the experiment ledger and are not erased from the scientific cost.

4. Architecture and capacity evidence

Model / comparison	Budget	Validation BPB	Interpretation
Course baseline, seed 17	1,200 steps	2.072081	Reference
RoPE only	1,200 steps	1.922026	Largest isolated gain
SwiGLU only	1,200 steps	2.011474	Useful but smaller gain
Modern bundle	1,200 steps	1.835657	Best first-screen result
RoPE+SwiGLU, 3-seed mean	1,200 steps	1.837202	Replicated simplified mechanism
Modern width 128/depth 4	4,800 steps	1.698374	Long equal-budget endpoint
RoPE+SwiGLU width 128/depth 4	4,800 steps	1.700166	Only 0.001791 behind modern
Modern width 256/depth 6	1,200 steps	1.690024	Best capacity-screen shape

The component ladder shows that not every modernizing change helps independently. RoPE supplies most of the early improvement; RoPE+SwiGLU nearly matches the full modern bundle at the long equal budget. The 0.001791 BPB difference between those two long runs is too small and single-seed to claim robust superiority. The full bundle was nevertheless retained for capacity search because it was the primary preregistered candidate, not because the small difference proved a mechanism.

Capacity trade-off

At 1,200 steps, width 192/depth 8 reaches 1.719857, width 224/depth 6 reaches 1.710101, and width 256/depth 6 reaches 1.690024 validation BPB. The largest candidate remains within all resource limits. Its unregularized 4,800-step run initially improves to 1.656612 at step 3,000 and then deteriorates, motivating a regularization screen rather than more unregularized training.

What this establishes

Equal-target evidence supports the claim that the architectural path improves predictive quality over the baseline. It does not isolate a statistically reliable contribution for every component. Capacity evidence supports a larger model within the allowed inference envelope, while the later dropout experiments address overfitting introduced by that capacity.

5. Regularization, duration, and replication

Setting	Steps	CPU FP32 validation BPB	Decision
Dropout 0.00	3,600	1.647222	Matched control
Dropout 0.05	3,600	1.602406	Large improvement
Dropout 0.10	3,600	1.597758	Retained winner
Dropout 0.15	3,600	1.624385	Rejected
Dropout 0.20	3,600	1.646837	Rejected
Dropout 0.10, fresh long schedule	4,800	1.575606	Adopt; gain 0.022153

Dropout 0.10 improves 0.049464 BPB over the matched no-dropout control. Stronger settings reverse the gain, providing a useful negative result and a bounded optimum rather than an open-ended sweep. The fresh 4,800-step schedule continues improving at every recorded validation point and passes the preregistered minimum gain of 0.002 BPB over the 3,600-step recipe.

Final-recipe learning curves

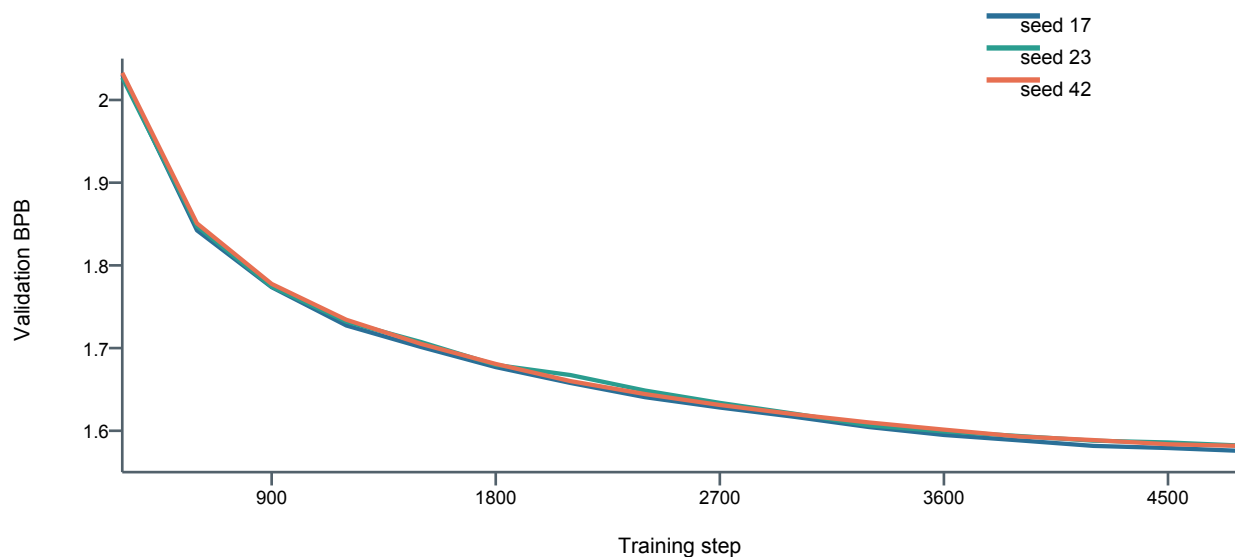


Figure 1. Complete validation BPB every 300 updates for the frozen recipe. All three seeds improve through the 4,800-step endpoint.

Seed	Selected step	CPU FP32 validation BPB	Checkpoint SHA-256 prefix
17	4800	1.575605888	ca3f431b8c90fa05
23	4800	1.582028099	c9072addfd1b2ea5
42	4800	1.581602162	6656baac0324155e
Mean	-	1.579745383	
Sample std.	-	0.003591228	

6. Resource compliance and final test

The formal pre-test resource gate launches a new process for each measurement and alternates baseline/candidate order across three repetitions. Peak RSS covers imports, model construction, fixed data loading, and scoring. The same frozen candidate used for this gate is the one later evaluated on test.

Metric	Observed	Limit	Result
Median CPU scoring time	43.163 s	5x baseline (12.384 s)	3.485x - pass
Maximum peak RSS	1,972,150,272 bytes	4,294,967,296 bytes	Pass
Core inference assets	21,136,388 bytes	67,108,864 bytes	Pass

The candidate full-test scoring field reports 93.19 seconds on this laptop, while the initial baseline test reports 25.79 seconds. These single test timings are consistent with the repeated validation ratio but are not substituted for the formal repeated gate.

Final score

1.605136467 full-test BPB
Protocol: 7506-mp1-wt2-v2 Targets: 428,405 UTF-8 bytes: 1,292,013

Predictor	Test BPB	Token PPL	Checkpoint SHA-256 prefix
Initial course baseline	2.102014912	80.975	1eb647dcae521f10
Frozen final model	1.605136467	28.658	ca3f431b8c90fa05

The frozen model lowers test BPB by 0.496878446, a 23.64% relative reduction. This result was observed only after the pre-test manifest recorded the exact source archive, model implementation, configuration, checkpoint, tokenizer, evaluator, selection evidence, and resource measurements.

7. Critical analysis

Strengths

- Controlled evidence: the baseline, primary architecture comparisons, and key ablations use equal processed-target budgets.
- Robustness: the final recipe is repeated at three seeds, and all three select the same endpoint behavior.
- Reproducibility: fixed files, sources, checkpoints, schedules, hashes, complete curves, negative experiments, and fresh-process resource runs are retained.
- Constraint awareness: capacity is increased only after a measured CPU/RAM gate; test is withheld until the predictor is frozen.

Limitations

- Three seeds quantify some initialization variance but do not establish a population-level guarantee or leaderboard rank.
- The final seed is selected by validation BPB across three runs. This is legitimate model selection but makes the selected 1.575606 validation value optimistic relative to the 1.579745 three-seed mean.
- Component effects are order-dependent. The modern bundle and simplified RoPE+SwiGLU model are nearly tied in one long comparison, so claims about RMSNorm and bias removal remain limited.
- Resource timing is machine-specific. Compliance is expressed as a same-machine ratio, while absolute seconds may differ on the instructor's CPU.
- Search explored a bounded set of architectures and regularization values. More compute might improve validation, but post-test changes are intentionally prohibited.

Failure analysis

The unregularized large model overfits after step 3,000, while dropout 0.15 and 0.20 underperform 0.10. These failures clarify the final choice: added capacity is useful only when paired with moderate regularization, and longer duration is beneficial only after regularization suppresses late-run deterioration. The experiments also show that adding every modern component is not automatically valuable; most early gain is attributable to RoPE.

No leaderboard guarantee

The measured test BPB is reproducible evidence for this frozen predictor, not a promise of rank. Rankings depend on other submissions and later verification. The model should therefore be judged by its fixed score, transparent controls, and reproducibility rather than speculative placement.

8. Reproduction, disclosure, and conclusion

Reproduction

Use Python 3.12 and the pinned dependencies in the repository. From code/, run the contract tests, then evaluate the supplied frozen checkpoint with the unchanged scorer:

Step	Command
Tests	<code>python -m unittest discover -s tests -v</code>
Fixed files	<code>python scripts/verify_fixed_files.py</code>
Final score	<code>python evaluate.py --checkpoint /path/checkpoint.pt --device cpu --precision fp32 --split test --output reproduced-test.json</code>

The expected BPB is 1.605136467. The authoritative checkpoint SHA-256 is ca3f431b8c90fa057f460604de8e9fcb4bf2c57f630ce99f7ed6af0bc63ac303. The frozen source archive SHA-256 is 3bf2e6c372c5df9175be323db428eb3c2b85deec9e2aaec4a02c6f35a3585e3.

AI assistance and responsibility

OpenAI Codex provided substantive assistance with assignment analysis, experiment framework and model code, tests, Windows deployment, staged experiment execution, result auditing, documentation, freezing, packaging, and this report draft. The student remains responsible for reviewing and understanding the implementation, checking all claims, acknowledging reused work, and ensuring that the final repository disclosure is accurate. Automated tests do not substitute for that human review.

Conclusion

A disciplined sequence of equal-budget ablation, capacity gating, regularization, duration testing, and three-seed replication reduces full-test BPB from 2.102015 to 1.605136 while remaining comfortably inside all inference limits. The strongest supported mechanism is RoPE, while SwiGLU and the larger regularized model contribute to the final score. The final predictor is frozen, hash-identified, directly evaluable without retraining, and accompanied by complete validation, resource, and test evidence.

References

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